

Employee Performance & Productivity Analytics

What 100,000 employee records reveal about pay, performance, attrition
and the HR myths the data dismantles.

Presented by **MOHAMMAD SHADAB**

Data Analyst · Python · SQL · Power BI · Excel

100,000 Employees

9 Departments

2014 – 2024

21 Charts · 4 Dashboard Tabs

What We'll Cover Today

01

Workforce Overview

How the company grew: headcount balance, the 2014–2024 hiring engine, tenure mix

02

Compensation

Job title vs education vs gender vs experience — what actually sets pay

03

Performance

Ratings, training and promotions — and the noise hiding inside them

04

Attrition & Satisfaction

Why 10,010 people left — and why satisfaction never warned us

If You Remember Only Three Things...

FINDING 01

Job title is the **ONLY** real pay lever

Engineers & Managers earn ~\$7.8K vs Technicians ~\$4.5K — a 72% gap. Education moves pay 0.52%, gender ~0.2%.

FINDING 02

Attrition is a flat 10% — everywhere

Department, education, remote work and even satisfaction barely move it. Stayers rate 3.00; leavers 2.99.

FINDING 03

Training & tenure don't drive results

Ratings are a uniform ~20K per score; training sits at 49.3–49.7 hrs at every rating. Correlations ≈ 0.00 .

\$6,403

AVG MONTHLY SALARY

10.01%

OVERALL ATTRITION

 $r = 0.51$ PERFORMANCE $\leftrightarrow$ SALARY**3.00 / 5**

AVG PERFORMANCE

THE TWIST

Almost every relationship in 100,000 rows is statistically zero — HR levers are disconnected from outcomes. Strategy should follow role architecture, not folklore.

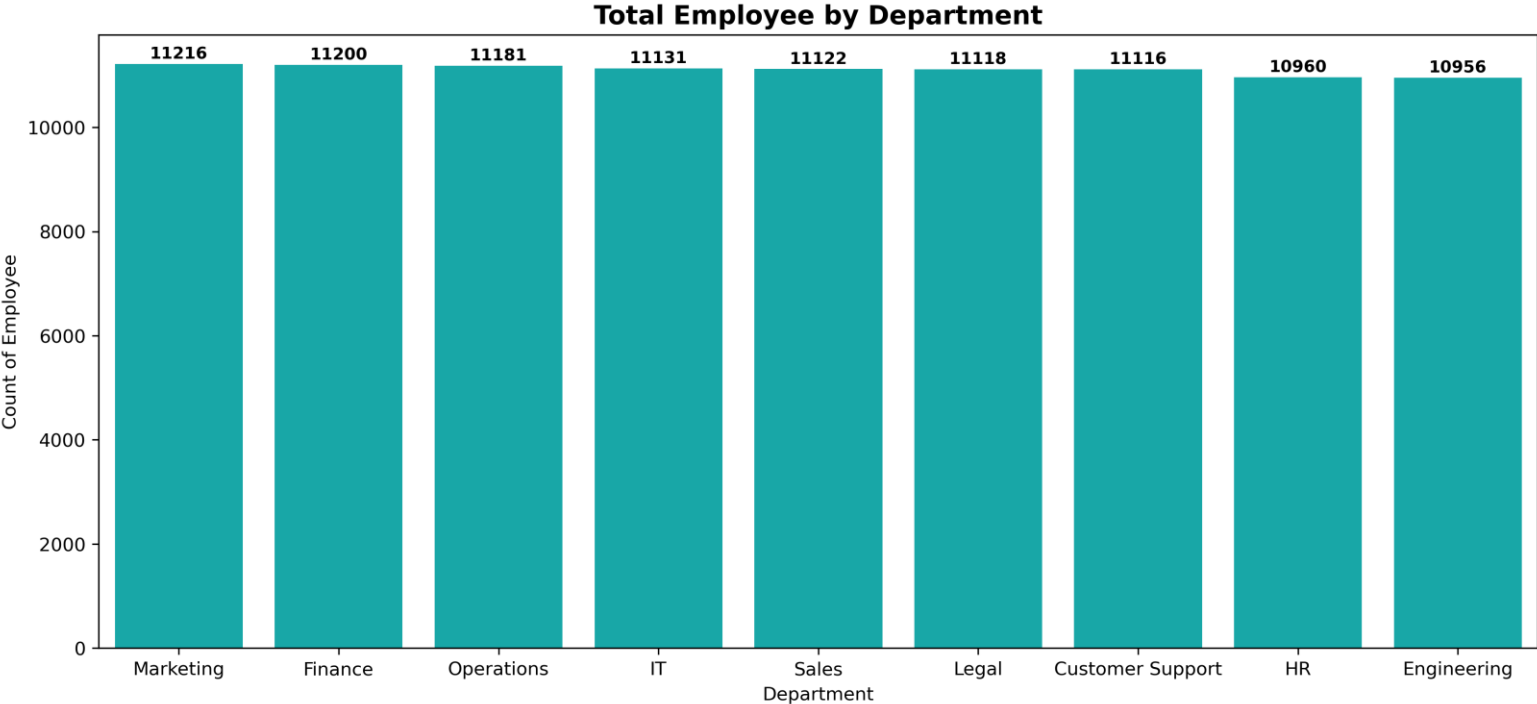
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SECTION ONE

Workforce Overview

How the company grew, where people sit, and what the hiring engine looked like over a decade.

Headcount by Department

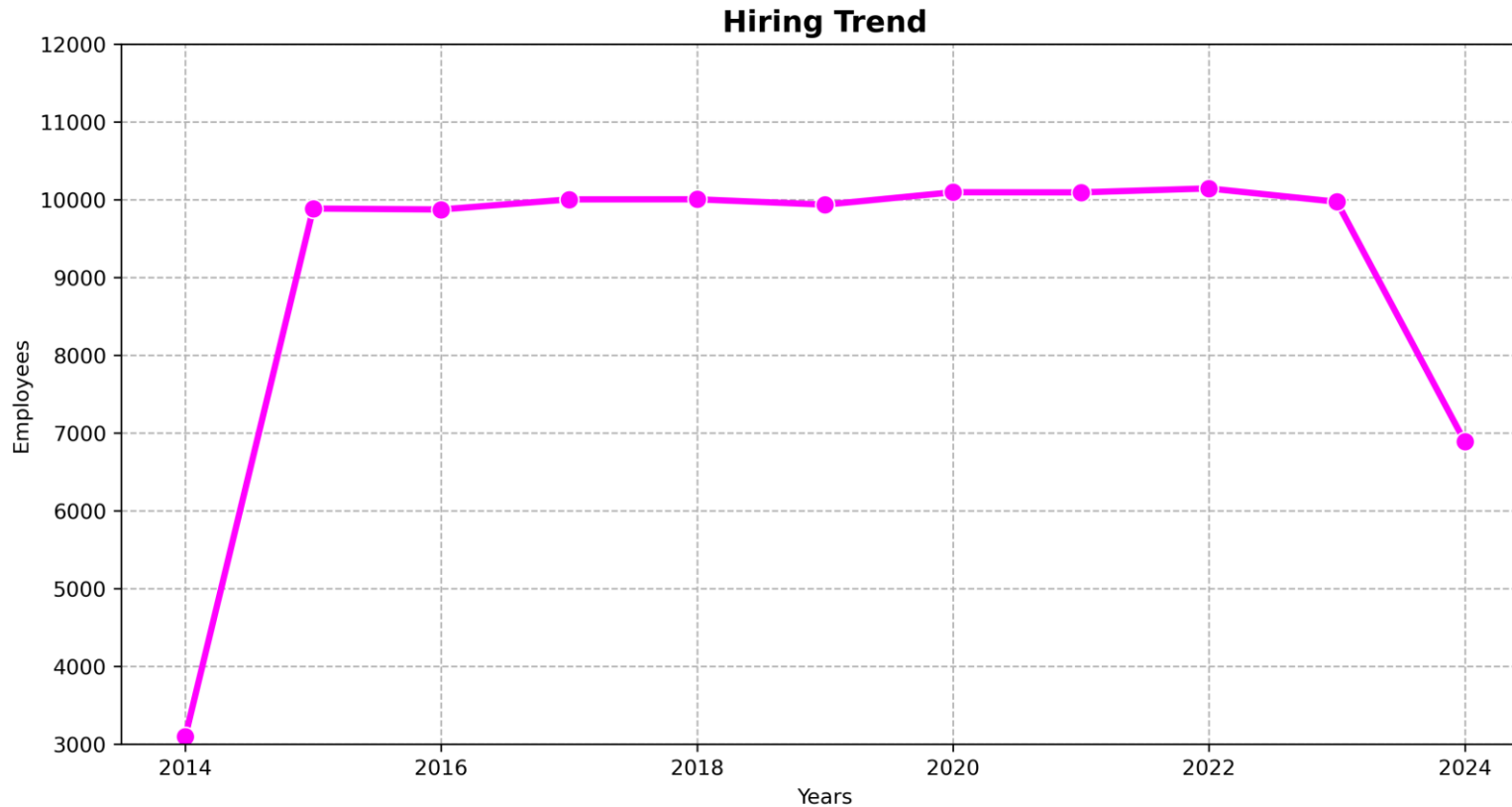


BALANCED BY DESIGN

Nine teams, one even spread

Marketing is the largest team (11,216); Engineering the smallest (10,956) — a difference of just 260 people (2.3%) across 100,000 employees. No department is staffed like a flagship or starved; workforce planning has been deliberately even-handed.

Hiring Trend, 2014–2024



RAMP · PLATEAU · SHOCK

The 2024 cliff is the story

2014→2015 headcount triples (3.1K→9.9K) in the build-out year; 2015–2023 is a stable plateau ~10K hires/year peaking at 10.1K in 2022. In 2024 hiring collapses to 6.9K — a 31% cut and the decade's only real shock, worth a post-mortem.

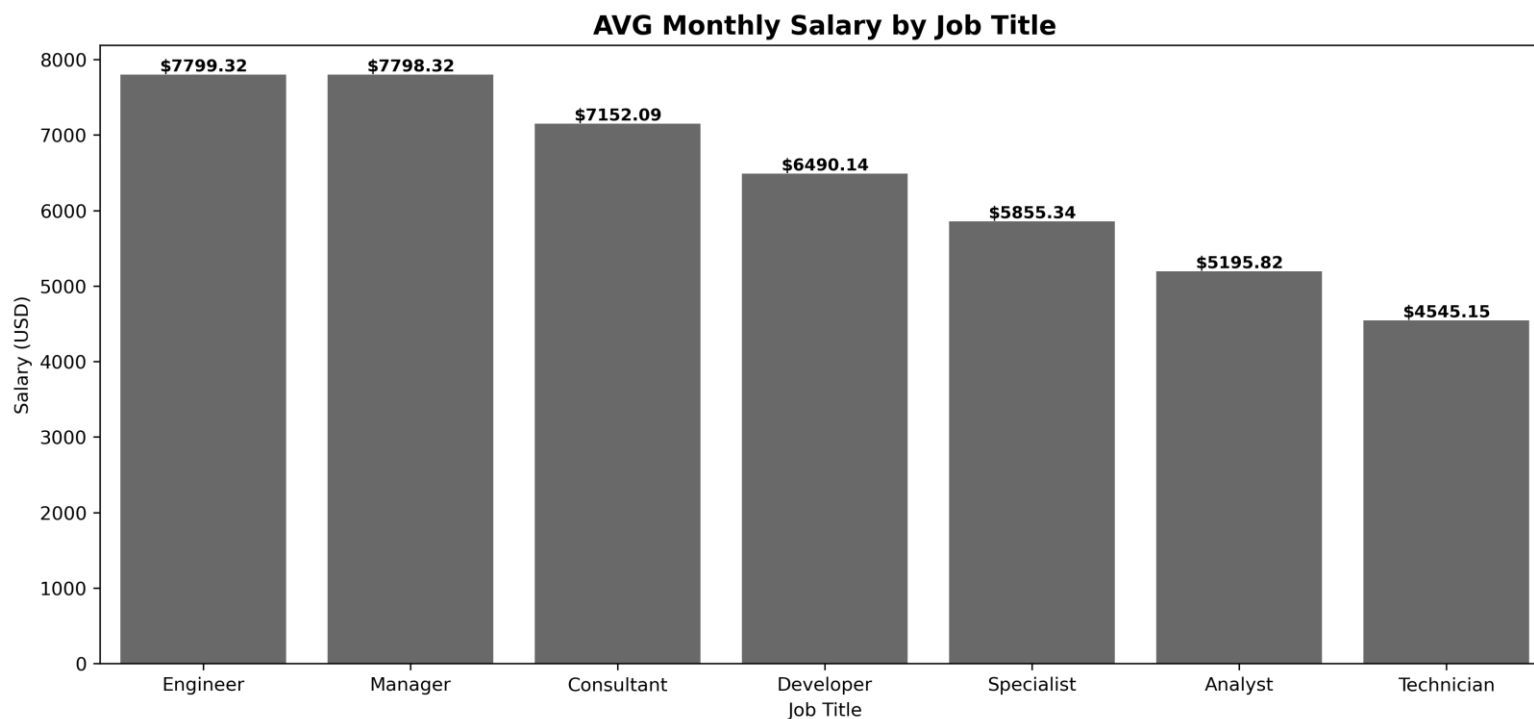
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SECTION TWO

Compensation

Who gets paid what — and why almost everything people THINK drives salary... doesn't.

Average Salary by Job Title — the 72% Gap



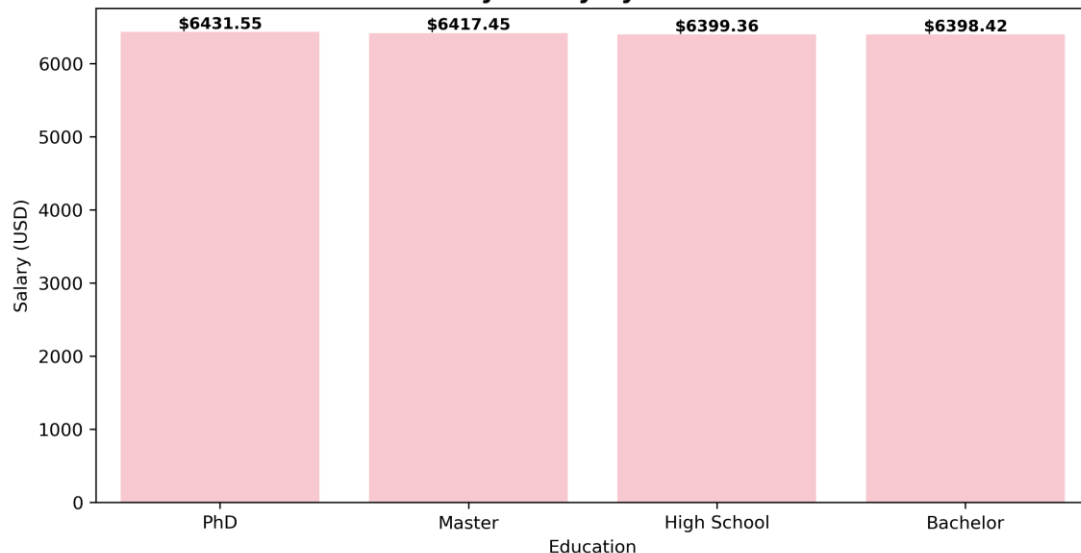
ROLE IS THE PAY SYSTEM

A \$3,254 monthly chasm

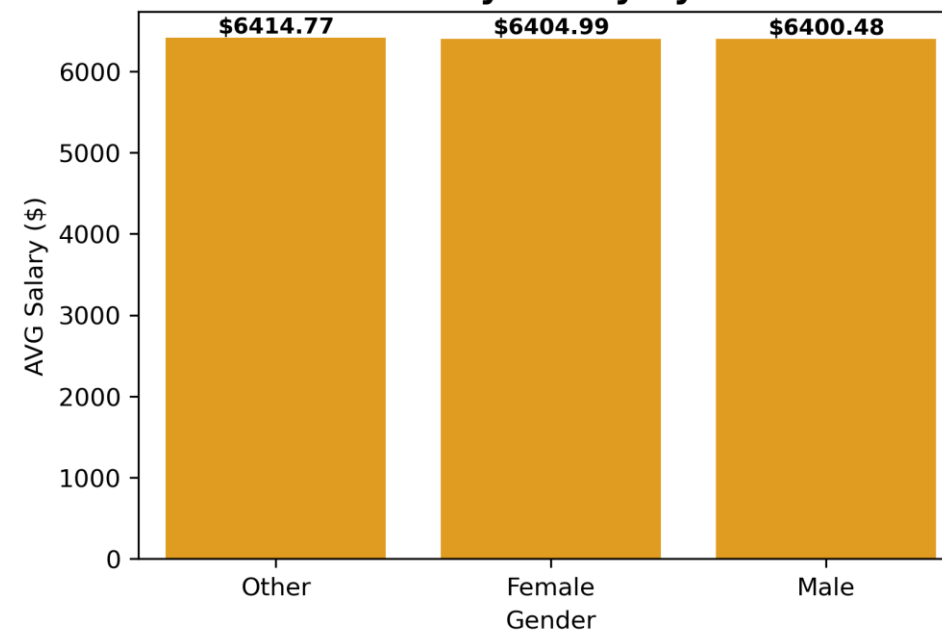
Engineers/Managers average \$7,799 while Technicians average \$4,545 — a 72% gap. Consultant (\$7,152) and Developer (\$6,490) sit mid-scale; Specialist (\$5,855) and Analyst (\$5,196) below. Job levelling is THE dial to turn for fairness and retention.

The Great Equalisers: Education & Gender

AVG Monthly Salary by Education Level



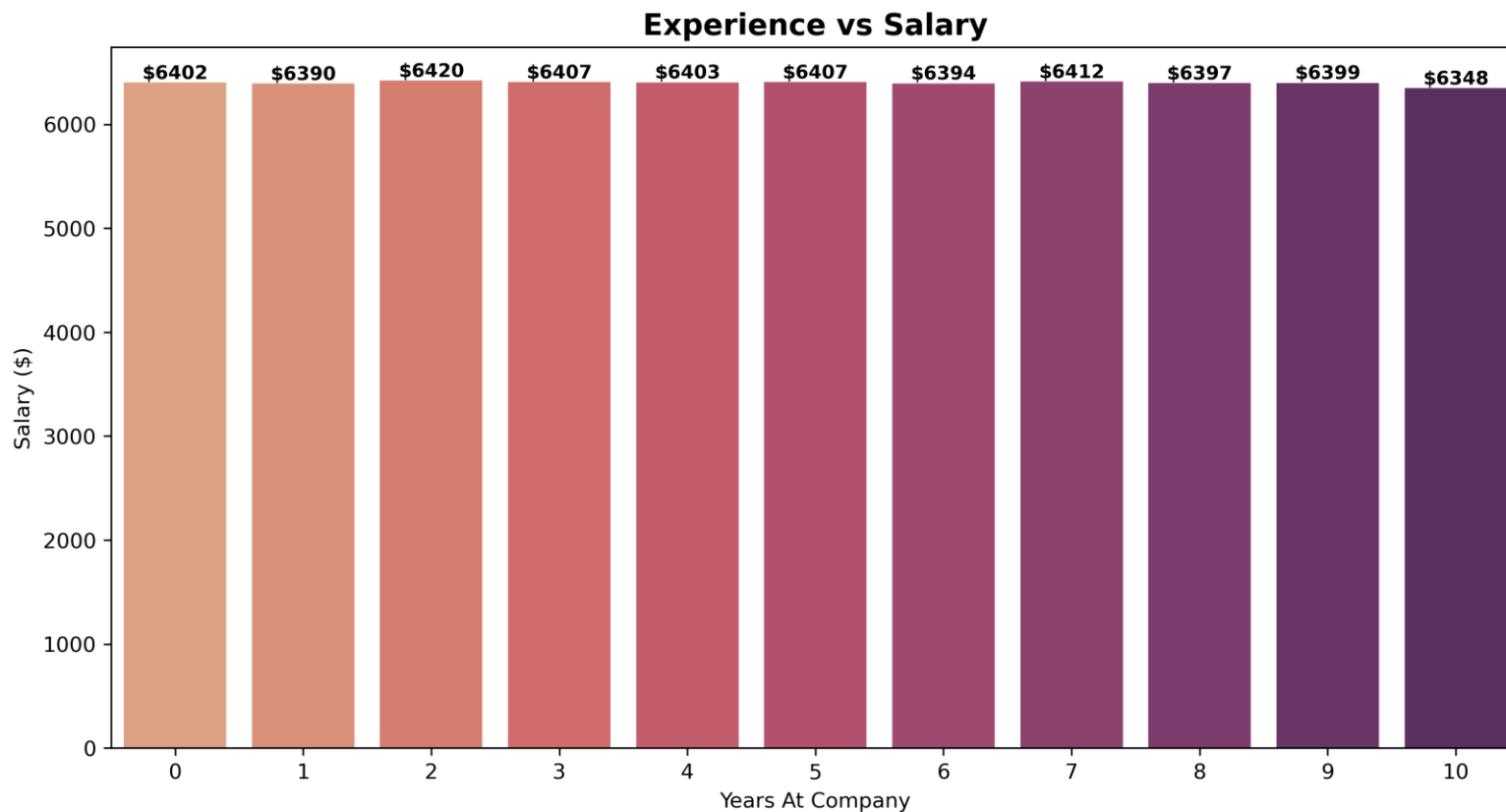
AVG Monthly Salary by Gender



MYTH BUSTED — AND A WIN TO PROTECT

A PhD earns \$6,431 vs a Bachelor's \$6,398 — about \$1/day more; education is table stakes, not a pay ladder. Gender pay gap $\approx 0.2\%$ (\$6,400–\$6,415 across groups): pay equity is essentially achieved — protect it in every audit.

Experience vs Salary: The Flatline



LOYALTY ISN'T PRICED IN

Year 10 earns the least

Year 0 earns \$6,402; year 10 earns \$6,348 — the most tenured cohort actually earns the LEAST, and the whole 11-year range is ~\$70 wide. Without time-based progression, long-tenure staff have every reason to look elsewhere; tenure increments are a cheap retention fix.

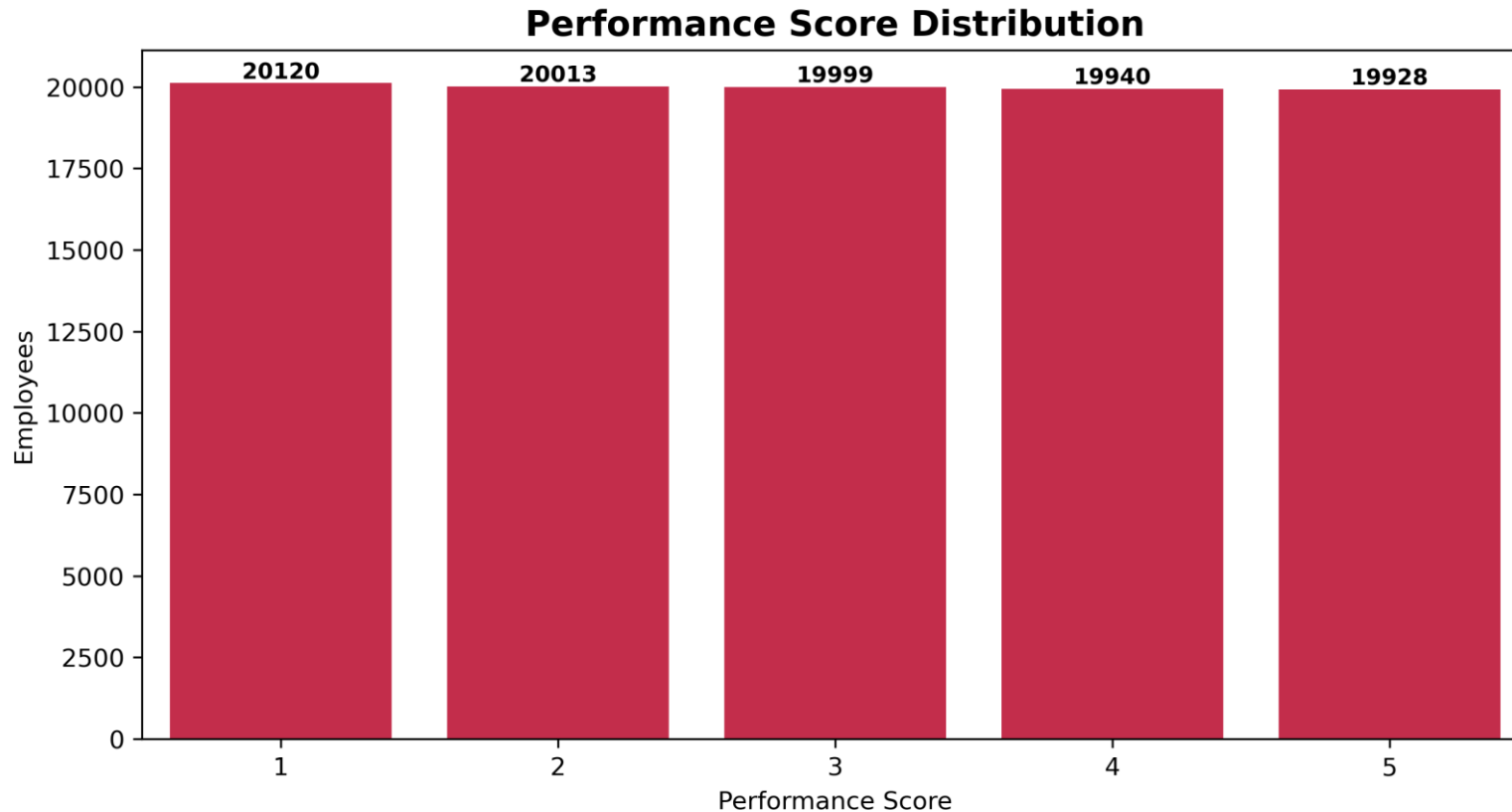
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SECTION THREE

Performance

Ratings, training and promotions — and why the performance machine looks suspiciously like a random number generator.

Performance Score Distribution

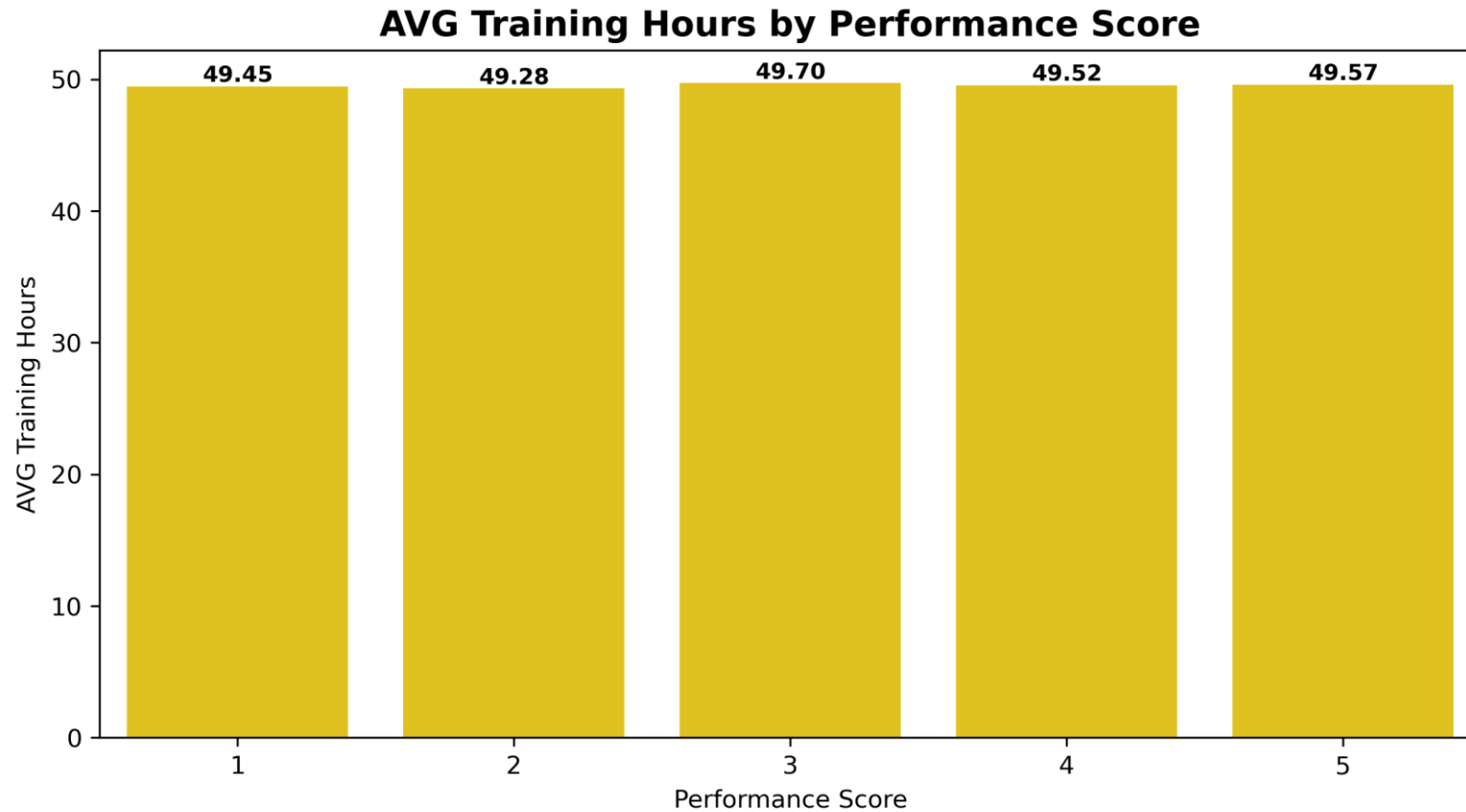


A WHITE-PICKET FENCE

20,000 people in every bucket

20,120 / 20,013 / 19,999 / 19,940 / 19,928 employees at scores 1–5 — just 192 people between the best and worst groups. Real reviews produce stars and laggards; a perfectly uniform spread says ratings carry almost no information.

Training Hours vs Performance Score

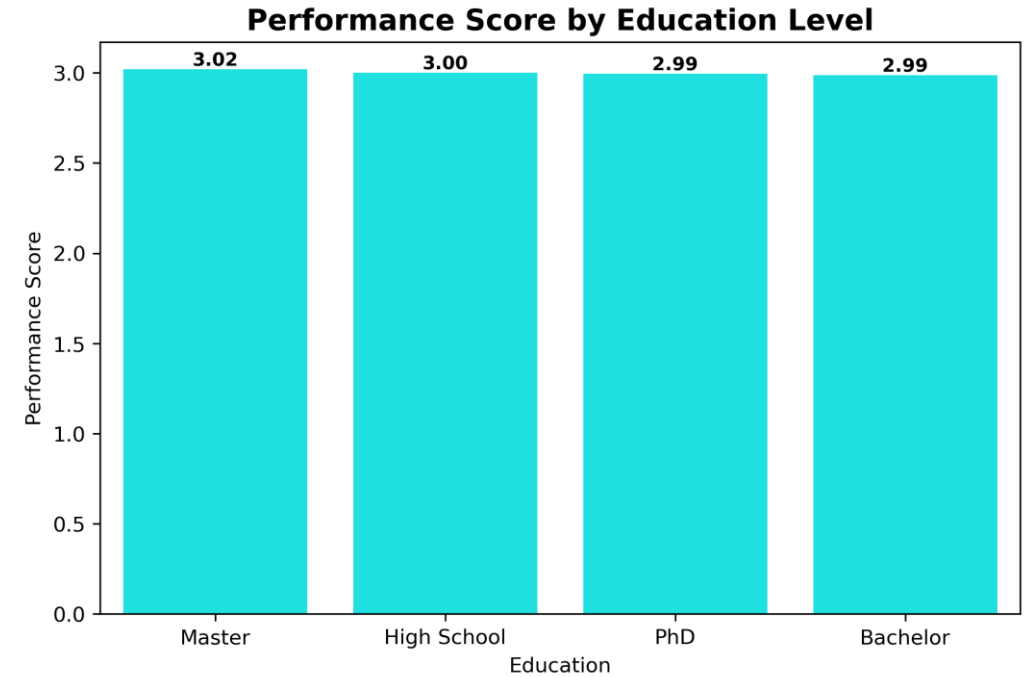
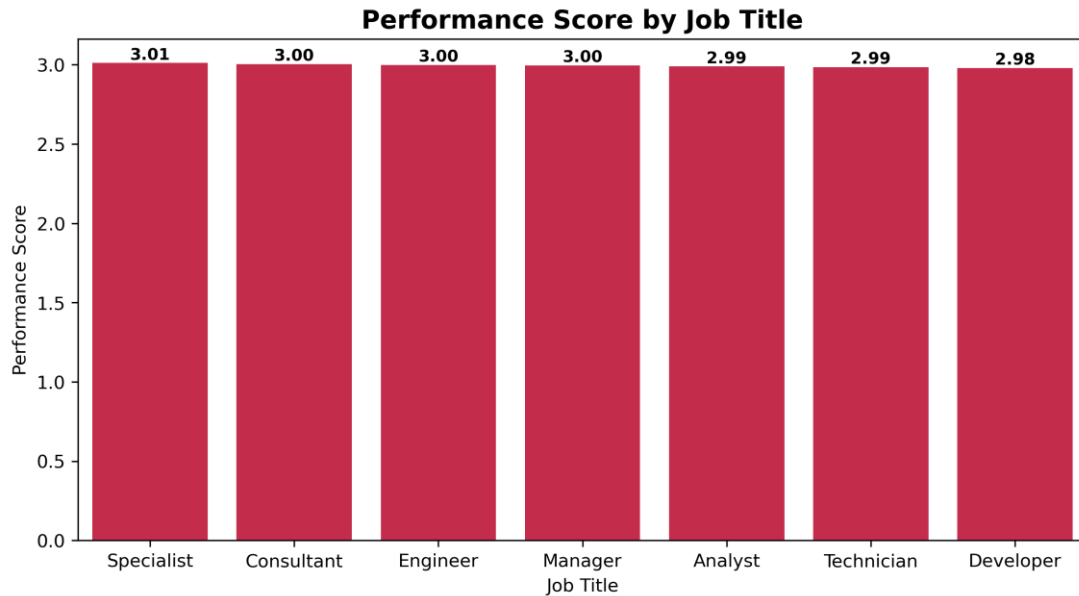


ENTITLEMENT, NOT INTERVENTION

A 25-minute range all year

Score 1 receives 49.45 training hours; score 5 receives 49.57 — a 25-MINUTE difference across a full year. Promotions don't move ratings either (3.01 with zero promotions vs 2.99 with 1–2). Training should be targeted at low performers and measured for lift.

Ratings by Job Title & Education — Everything Is a 3.00



UNDIFFERENTIATED ACROSS THE BOARD

Every job title scores 2.98–3.01, every education level 2.99–3.02, every gender 2.99–3.00. Yet Performance $\leftrightarrow$ Salary is the dataset's ONLY real correlation ($r = 0.51$): uninformative ratings are driving pay. Reviews need evidence anchors and manager calibration.

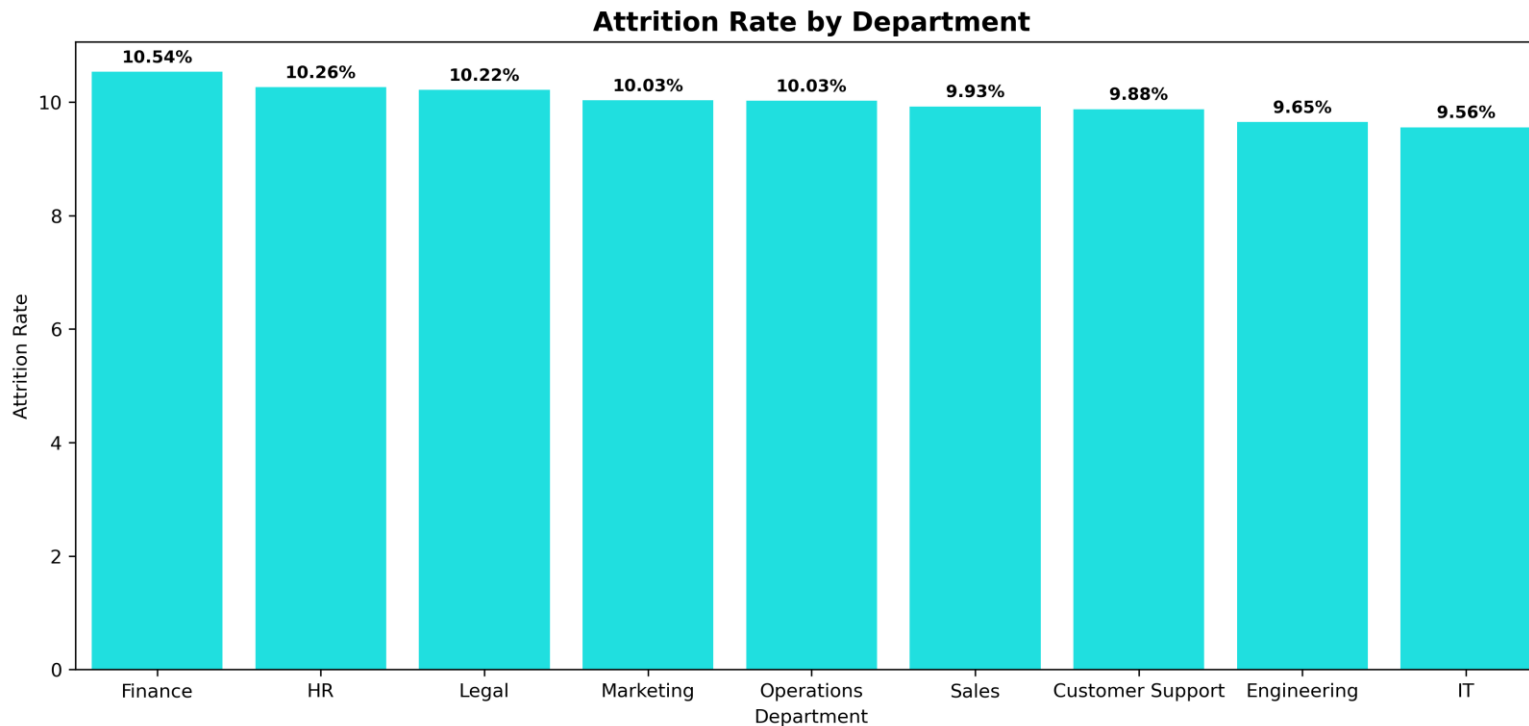
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SECTION FOUR

Attrition & Satisfaction

10,010 people left. The data can barely tell them apart from the 89,990 who stayed — and that's the most important finding of all.

Attrition Rate by Department

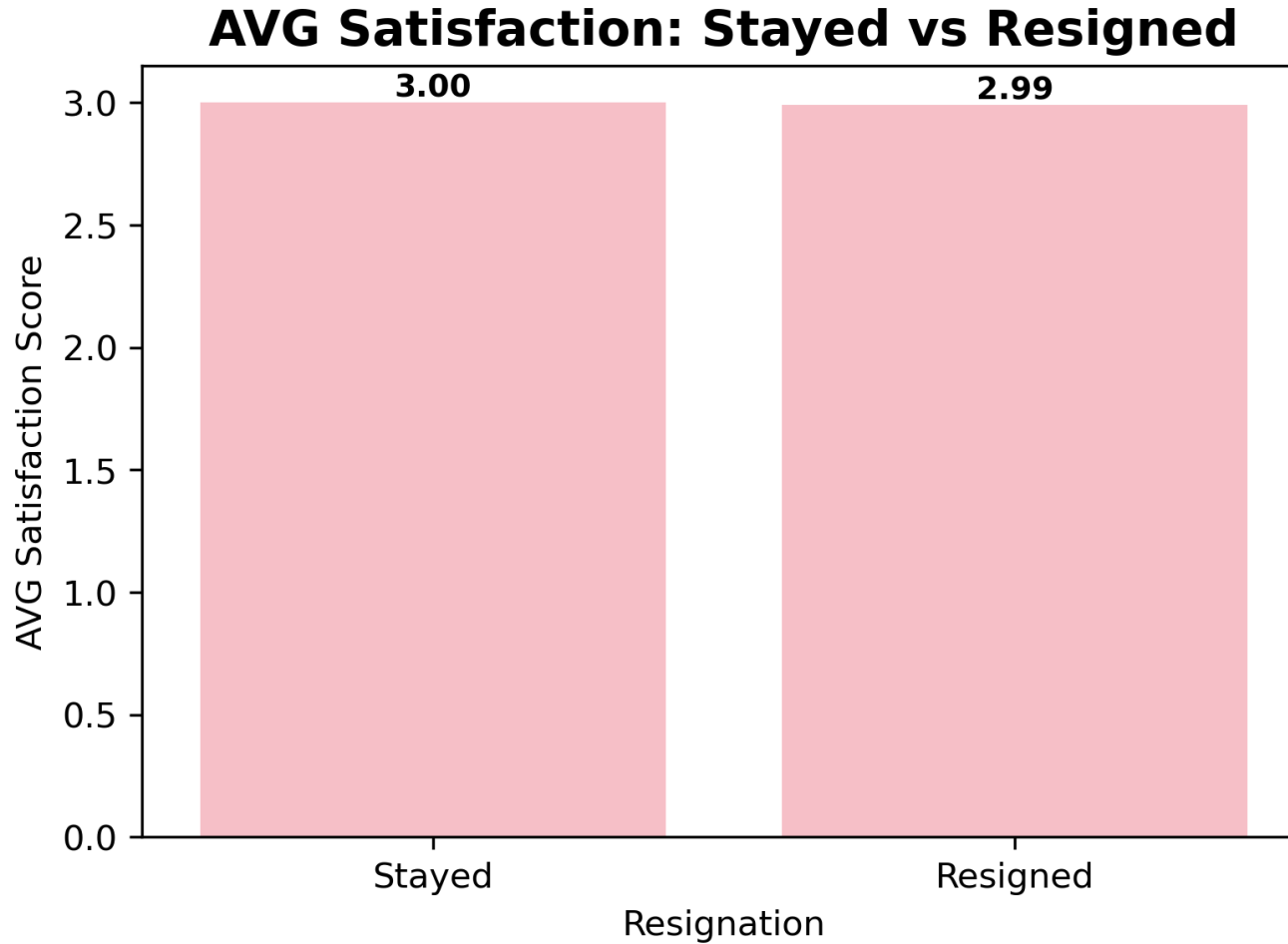


A COMPANY-WIDE CONSTANT

Turnover isn't local

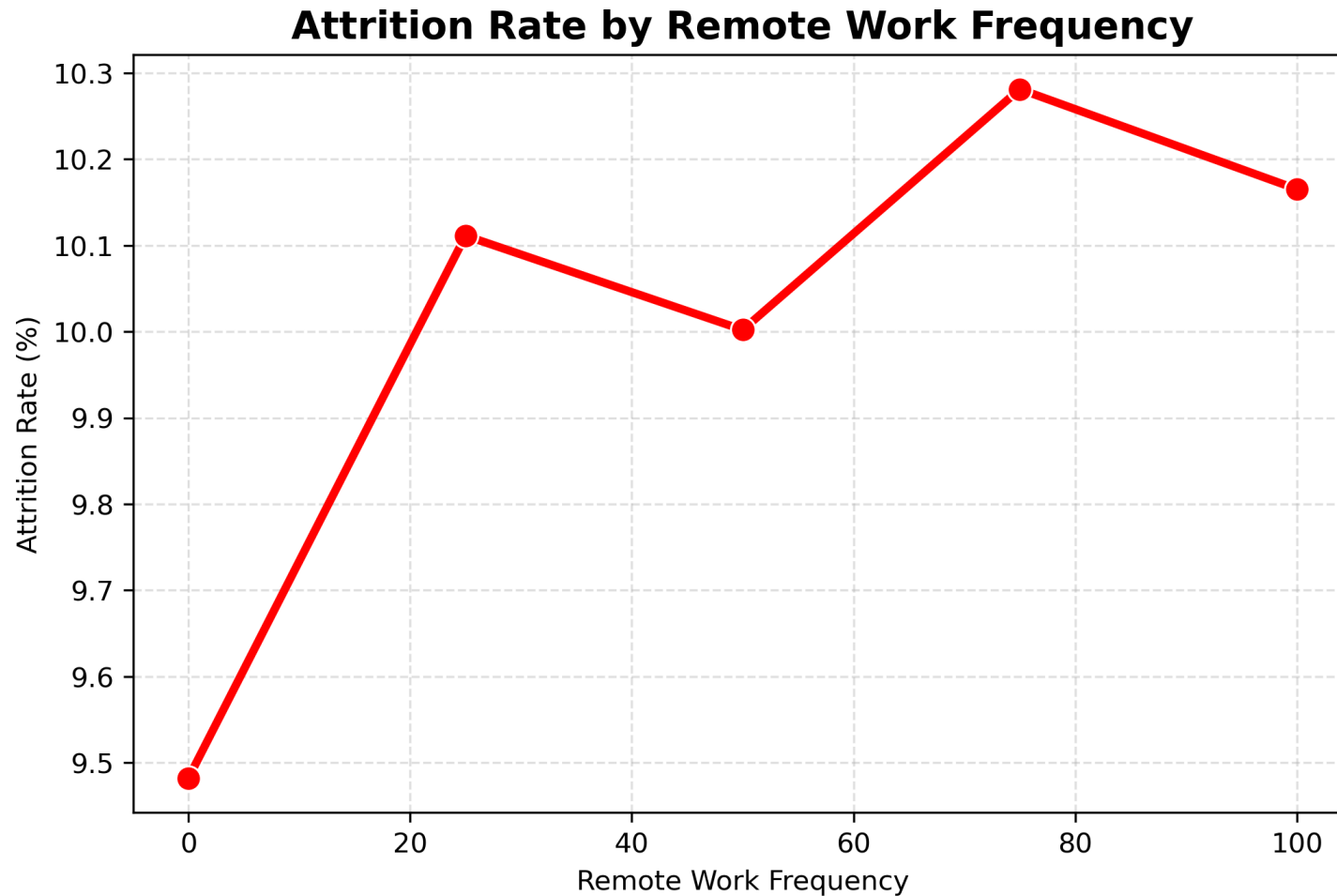
Finance leads at 10.54%, IT is lowest at 9.56% — the entire 9-department range is under 1 percentage point. Education is flat too (PhD 10.31% → Master 9.98%). Turnover points to systemic causes, not 'bad managers' — local fixes won't move a global number.

The Satisfaction Paradox

**THE EXIT SIGNAL IS MISSING****Stayed 3.00 vs Resigned 2.99**

A 0.01 difference on a 5-point scale: the people who walked out were statistically exactly as happy as those who stayed. Satisfaction surveys predict ZERO attrition. Replace passive mood scoring with intent-to-stay pulses, manager-quality signals and compa-ratio flags.

Attrition vs Remote Work Frequency

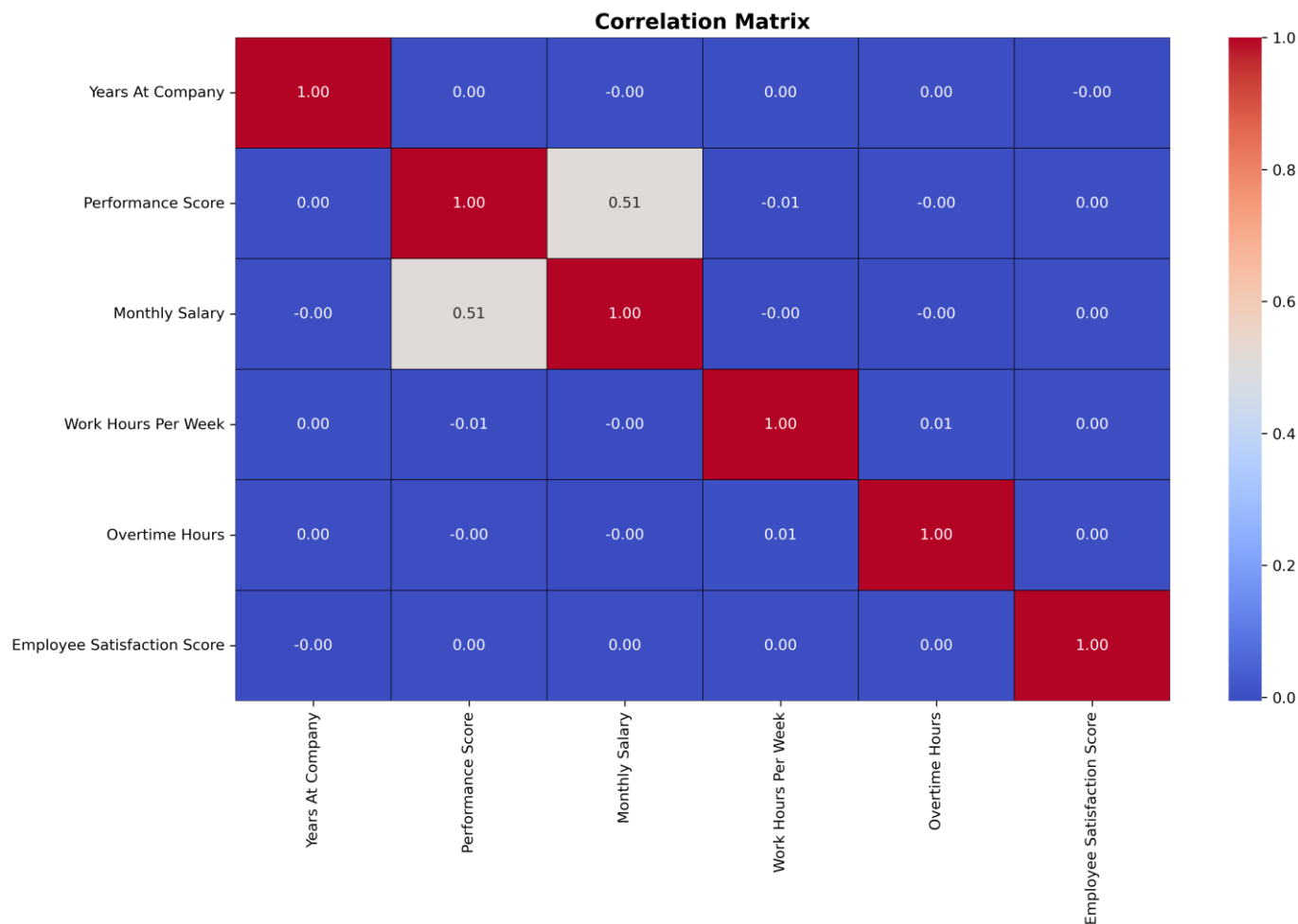


A TUNING KNOB, NOT A DRIVER

0.8 points from end to end

Fully in-office (0%) = 9.48% attrition; 75% remote = 10.28% peak; fully remote = 10.17%. The whole curve spans just 0.8 points — remote policy isn't the cause. But paired with the 2024 hiring cliff, hybrid-heavy teams deserve a closer qualitative look.

Correlation Matrix: One Signal in a Sea of Zeroes



R = 0.51 IS THE WHOLE GAME

One relationship survives

Across 15 numeric features, Performance ↔ Salary (0.51) is the only correlation that isn't noise. Training ↔ performance, tenure ↔ salary, satisfaction ↔ attrition and hours ↔ overtime all round to 0.00.

A dashboard that finds 'nothing relates' is not a failed analysis — it's a diagnosis: people processes are running unmeasured. That's where strategy starts.

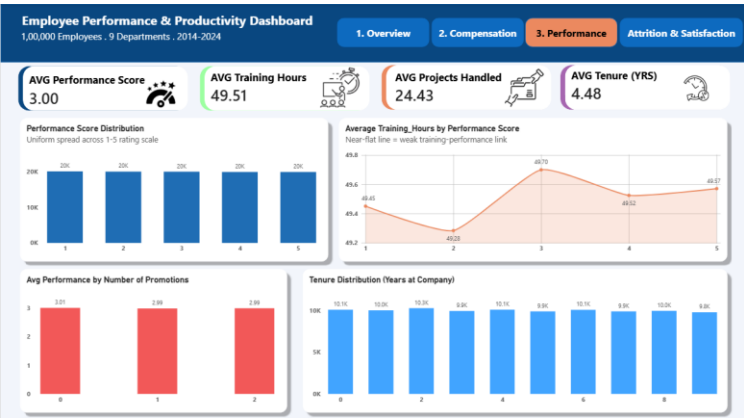
Interactive Dashboard — Four Tabs



Tab 1 · Overview — headcount, hiring, demographics



Tab 2 · Compensation — pay across role, education, gender



Tab 3 · Performance — ratings, training, tenure



Tab 4 · Attrition & Satisfaction — turnover drivers

What HR Believes vs What 100,000 Records Show

The belief	The evidence	Verdict
Higher education = higher pay	PhD \$6,431 vs Bachelor \$6,398	BUSTED · 0.52%
More experience = higher pay	Year 0 \$6,402 · Year 10 \$6,348	BUSTED · flat
Training boosts performance	49.28–49.70 hrs at every rating	BUSTED · 25-min range
Unhappy people quit	Stayed 3.00 vs Resigned 2.99	BUSTED · 0.01
Remote workers leave more	9.48% in-office vs 10.28% at 75%	MOSTLY BUSTED · 0.8 pt
Ratings differentiate people	~20,000 employees in every score	BUSTED · uniform
Job title drives pay	\$4,545 Technician → \$7,799 Engineer	CONFIRMED · 72%
Pay is gender-equitable	\$6,400–\$6,415 across all groups	CONFIRMED · 0.2%

Six Recommendations for Leadership

1

Rebuild job levelling; publish bands

The 72% title gap is the whole pay story. Create visible paths for Technicians & Analysts stuck below \$5.9K.

2

Add tenure-based progression

Year-10 staff earn the least (\$6,348). Time-in-role increments beat replacing institutional knowledge.

3

Rebuild the churn early-warning system

Satisfaction predicts nothing. Add intent-to-stay pulses, manager NPS, compa-ratio flags + a leavers model.

4

Target training; stop rationing it equally

Aim hours at low performers and measure rating lift vs a control group.

5

Calibrate performance reviews

Uniform ratings drive pay ($r = 0.51$) yet say nothing. Use evidence anchors + calibration sessions.

6

Investigate the 2024 signal

31% hiring drop + peak hybrid attrition — run a post-mortem starting with Finance (10.54%).



Thank You

Questions & Discussion

The best dashboard doesn't just show data — it challenges beliefs.
One real lever (role), one win (pay equity), five metrics to rebuild.

Mohammad Shadab · Data Analyst

Python · SQL · Power BI · Excel

100,000 Employees · 9 Departments · 2014–2024